

## Model Development Phase Template

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|---------------|----------------------------------------------------------------|
| Date          | 02 Feb 2026                                                    |
| Team ID       | LTVIP2026TMIDS75146                                            |
| Project Title | Rising Waters: A Machine Learning Approach To Flood Prediction |
| Maximum Marks | 6 Marks                                                        |

### Model Selection Report

In the forthcoming Model Selection Report, various models will be outlined, detailing their descriptions, hyperparameters, and performance metrics, including Accuracy or F1 Score. This comprehensive report will provide insights into the chosen models and their effectiveness.

### Model Selection Report:

| Model         | Description                                                                                                                                                                                             | Hyperparameters | Performance Metric (e.g., Accuracy, F1 Score) |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------|
| Decision Tree | An ensemble of decision trees that aggregates results to boost accuracy and reduce overfitting. In this project, it achieved 100% accuracy and demonstrated strong generalization for flood prediction. | -               | Accuracy score: 100%                          |

|               |                                                                                                                                                                                                                |   |                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------|
| Random Forest | A simple, interpretable model that splits data based on feature thresholds. It provided 100% accuracy but may be prone to overfitting without pruning or tuning. Suitable for understanding feature influence. | - | Accuracy score: 100%         |
| KNN           | A distance-based classifier that assigns labels based on the nearest neighbors. It achieved 91.3% accuracy, performing well overall but showed reduced recall for floodpositive cases (1 class).               | - | Accuracy score: 91.3%<br>KNN |
| XGB           | An advanced gradient boosting model that builds trees sequentially to optimize prediction accuracy. It handled complex relationships effectively, achieving 91.3% accuracy on this dataset.                    | - | Accuracy score: 91.3%<br>XGB |